

Finite Box Dimension Forces Finite-Rank Taylor Coefficients

Automated mathematics run `fa_banach.001`, lane 10

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Status. Candidate full affirmative solution to Questions 4.4 and 4.5 of Carando–D’Andrea–Torres–Turco, likely valid, pending human review. The proof gives the stronger conclusion that every homogeneous Taylor coefficient has finite-dimensional range, with an explicit rank bound.

1 Source questions and answer

Let E, F be complex Banach spaces, let $U \subset E$ be open, and let $f : U \rightarrow F$ be holomorphic. Write

$$f(x_0 + x) = \sum_{m=0}^{\infty} P_m(x)$$

for its Taylor expansion at x_0 , where $P_m = P_m f(x_0)$ is m -homogeneous. Questions 4.4 and 4.5 of [1] ask whether

$$\dim_B f(x_0 + \varepsilon B_E) < \infty$$

forces, for every m , respectively

$$(e_n(P_m(B_E)))_{n \geq 1} \in \ell_1 \quad \text{and} \quad \overline{\dim}_B P_m(B_E) < \infty.$$

The printed questions write $f : U \rightarrow E$ although F has just been introduced; as everywhere else in the paper, the well-typed formulation is $f : U \rightarrow F$.

The answer to both questions is *yes*. In fact, if

$$d := \overline{\dim}_B f(x_0 + \varepsilon B_E) < \infty, \quad r := \left\lfloor \frac{d}{2} \right\rfloor,$$

and

$$V_m := \text{span}(P_1(E) \cup \cdots \cup P_m(E)), \quad V_0 = \{0\},$$

then

$$\dim(V_m/V_{m-1}) \leq (r+m)^m, \quad \dim V_m \leq \sum_{k=1}^m (r+k)^k. \tag{1}$$

Since we are assuming that $\overline{\dim}_B f(B_E) < \infty$, by Proposition 1.2 we have that $\limsup_{n \rightarrow \infty} \sqrt[n]{e_n(f(B_E))} < 1$. This and (8) complete the proof. $\square$

We do not know if Proposition 4.2 holds for $p = 1$. More precisely, we have the following question

Question 4.4. Let E and F be Banach spaces, $U \subset E$ an open set, and $x_0 \in U$. Take $f: U \rightarrow E$ an holomorphic function and suppose that there exists $\varepsilon > 0$ such that $\dim_B f(x_0 + \varepsilon B_E) < \infty$. Is it true that $(e_n(P_m f(x_0)(B_E)))_n$ belongs to ℓ_1 for all $m \in \mathbb{N}$?

A probably more natural question, whose positive answer would clearly imply a positive answer to Question 4.4, is the following.

Question 4.5. Let E and F be Banach spaces, $U \subset E$ an open set, and $x_0 \in U$. Take $f: U \rightarrow E$ an holomorphic function and suppose that there exists $\varepsilon > 0$ such that $\dim_B f(x_0 + \varepsilon B_E) < \infty$. Is it true that for every $m \in \mathbb{N}$, $\overline{\dim}_B P_m f(x_0)(B_E) < \infty$?

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Figure 1: Official arXiv:2401.12059v1 PDF page 15: the end of Proposition 4.2 and Questions 4.4–4.5.

2 Two rank lemmas

The first observation converts metric dimension into a uniform differential rank bound. Complex rank contributes twice its value to real box dimension.

Lemma 1 (box dimension bounds differential rank). *Put $K = f(x_0 + \varepsilon B_E)$ and suppose $d = \overline{\dim}_B K < \infty$. Then*

$$\text{rank}_{\mathbb{C}} Df(y) \leq \lfloor d/2 \rfloor \quad (y \in x_0 + \varepsilon B_E).$$

Proof. Suppose $Df(y)$ has rank at least s . Choose $h_1, \dots, h_s \in E$ and $\lambda_1, \dots, \lambda_s \in F^*$ so that the matrix

$$(\lambda_i(Df(y)h_j))_{i,j=1}^s$$

is invertible. On $H = \text{span}\{h_1, \dots, h_s\}$ consider

$$\Phi(z) = (\lambda_1(f(y+z)), \dots, \lambda_s(f(y+z))).$$

After shrinking a neighborhood of 0 in H , it stays inside the given ball. The finite-dimensional holomorphic inverse theorem says that its image contains a nonempty open subset of $\mathbb{C}^s$. Hence the Lipschitz image $(\lambda_1, \dots, \lambda_s)(K)$ has upper box dimension at least $2s$. Lipschitz maps do not increase upper box dimension, so $2s \leq d$. Taking $s = \lfloor d/2 \rfloor + 1$ proves the claim. $\square$

The second lemma is the Banach-space form of the Jacobian estimate proved in Section 3 of [1]. We include the reduction because it is exactly what the Taylor argument needs.

Lemma 2 (bounded derivative rank forces finite polynomial rank). *Let $P : E \rightarrow G$ be a continuous m -homogeneous polynomial and suppose*

$$\text{rank } DP(x) \leq r \quad (x \in E).$$

Then

$$\dim \text{span } P(E) \leq (r + m)^m.$$

Proof. If $P(x_1), \dots, P(x_N)$ are linearly independent, choose $\lambda_1, \dots, \lambda_N \in G^*$ with $\lambda_i(P(x_j)) = \delta_{ij}$. Define $L : \mathbb{C}^N \rightarrow E$ by $L(a) = \sum_j a_j x_j$ and define the scalar m -homogeneous polynomials

$$p_i = \lambda_i \circ P \circ L \quad (1 \leq i \leq N).$$

They are linearly independent because $p_i(e_j) = \delta_{ij}$. The Jacobian-rank estimate of [1, Corollary 3.3] gives

$$\max_{a \in \mathbb{C}^N} \text{rank } D(p_1, \dots, p_N)(a) \geq N^{1/m} - m.$$

On the other hand, each of these differentials factors through $DP(L(a))$, so its rank is at most r . Thus $N \leq (r + m)^m$. If $\text{span } P(E)$ had larger dimension, one could choose more than $(r + m)^m$ independent values, a contradiction. $\square$

For completeness, the estimate invoked above comes from a triangular coefficient count. If the maximum Jacobian rank of N linearly independent m -homogeneous scalar polynomials is q , choose a nonzero q -minor. Every polynomial in their span satisfies the associated first-order determinantal equations. After a generic linear change of variables, coefficient comparison in lexicographic order shows that a degree- m solution is determined by the monomials in the first q variables. There are $\binom{q+m-1}{m} \leq (q+m)^m$ of these. Hence $N \leq (q+m)^m$, which is precisely the displayed lower bound. This also clarifies that only the final estimate of Corollary 3.3 is being used; no infinite-dimensional algebraic geometry is hidden in Lemma 2.

3 Taylor filtration and the noncancellation determinant

Theorem 3 (quantitative affirmative answer). *Under the hypotheses in Section 1, assume only that $d = \overline{\dim}_B f(x_0 + \varepsilon B_E) < \infty$. Then every V_m is finite dimensional and the bounds (1) hold. In particular, each P_m has finite-dimensional range and $P_m(B_E)$ has finite upper box dimension.*

Proof. Set $r = \lfloor d/2 \rfloor$. We argue by induction on m . Suppose V_{m-1} is finite dimensional. It is then closed, so the quotient $G_m = F/V_{m-1}$ is a Banach space. Let

$$\pi_m : F \rightarrow G_m, \quad Q_k = \pi_m \circ P_k, \quad g = \pi_m \circ f.$$

By construction, $Q_k = 0$ for $k < m$.

We claim that

$$\text{rank } DQ_m(x) \leq r \quad (x \in E). \quad (2)$$

If not, put $s = r + 1$ and choose $h_1, \dots, h_s \in E$ and $\mu_1, \dots, \mu_s \in G_m^*$ such that

$$M = (\mu_i(DQ_m(x)h_j))_{i,j=1}^s \quad \text{has} \quad \det M \neq 0.$$

For sufficiently small t , the point $x_0 + tx$ lies in the prescribed ball, and termwise differentiation of the Taylor series gives

$$Dg(x_0 + tx)h = \sum_{k=m}^{\infty} t^{k-1} DQ_k(x)h.$$

Therefore the analytic scalar function

$$\Delta(t) = \det(\mu_i(Dg(x_0 + tx)h_j))_{i,j=1}^s$$

satisfies

$$\Delta(t) = t^{s(m-1)} \det M + O(t^{s(m-1)+1}). \quad (3)$$

It is nonzero for all sufficiently small nonzero t . Consequently $Dg(x_0 + tx)$ has rank at least s . Since $Dg = \pi_m \circ Df$, this forces $Df(x_0 + tx)$ to have rank at least $s = r + 1$, contradicting Lemma 1. This proves (2).

Lemma 2, applied to Q_m , now gives

$$\dim \text{span } Q_m(E) \leq (r + m)^m.$$

But $\text{span } Q_m(E)$ is canonically V_m/V_{m-1} . This proves the first bound in (1); summing the successive dimensions proves the second and closes the induction.

Finally $P_m(B_E)$ is a bounded subset of the finite-dimensional complex space $\text{span } P_m(E) \subseteq V_m$. Its upper box dimension is therefore finite (indeed, at most $2 \dim \text{span } P_m(E)$). $\square$

The quotient in this proof is essential. Without it, an $s \times s$ minor of $Df(x_0 + tx)$ may contain lower Taylor degrees, and its coefficient at order $s(m-1)$ can cancel the minor coming from $DP_m(x)$. Passing to F/V_{m-1} removes every lower degree and makes the leading coefficient in (3) exact.

Corollary 4 (the missing ℓ_1 endpoint). *For every $m \geq 1$,*

$$(e_n(P_m(B_E)))_{n \geq 1} \in \ell_1.$$

Thus Questions 4.4 and 4.5 both have affirmative answers.

Proof. Let $D_m = \dim \text{span } P_m(E) < \infty$. If $D_m = 0$ there is nothing to prove. Otherwise the standard volumetric covering estimate in the real $2D_m$ -dimensional normed space gives, for a constant C_m ,

$$e_n(P_m(B_E)) \leq C_m 2^{-n/(2D_m)}.$$

The right-hand side is summable. $\square$

4 Verification, novelty, and review focus

The exact checker bundled with this packet verifies, over the integers, the leading exponent and leading determinant in (3) for matrix sizes 1 through 4 and Taylor degrees 1 through 5. It also checks the cumulative bounds for representative box dimensions. These computations guard the exponent and indexing; the general proof is the determinant argument above.

Six focused attempts are recorded in the run note. Direct entropy and convex hull methods reproduce the source’s $p > 1$ loss; diagonal counterexample families cannot place infinite rank in one fixed homogeneous degree. The successful route instead uses differential rank, lower-degree quotients, and the source’s algebraic Jacobian estimate.

A bounded literature search on 13 August 2026 covered the exact title, Questions 4.4–4.5, arXiv id, DOI, authors, and combinations of “finite box dimension”, “Taylor coefficient”, “finite rank”, and “entropy endpoint”. It found the 2025 Mathematische Nachrichten publication and one indexed citing paper on generalized entropy numbers, but no claimed answer to either question. Novelty confidence is moderate because citation databases and web search are not exhaustive.

Human review should prioritize Lemma 2’s reduction to Corollary 3.3 of the source and the quotient determinant in (3). The inverse-function and finite-dimensional entropy steps are standard. The argument answers a slightly stronger hypothesis than printed: finite *upper* box dimension suffices, whether or not the box-dimension limit exists.

References

- [1] D. Carando, C. D’Andrea, L. A. Torres, and P. Turco, *Entropy numbers and box dimension of polynomials and holomorphic functions*, Mathematische Nachrichten 298 (2025), 567–580; arXiv:2401.12059v1.